

Customer Shopping Behavior Analysis

Prepared by Preet | Retail Customer & Sales Analytics

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across four product categories. A retail company wants to understand which factors — discounts, reviews, seasons, and payment preferences — actually drive purchase decisions and repeat purchases, in order to improve sales, customer engagement, and marketing strategy. This report walks through the data preparation, the SQL analysis performed to answer that question, the dashboard built to communicate it, and the recommendations that follow from the findings.

2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18 (17 after cleaning — a redundant column was dropped)
- **Key Features:**
 1. Customer demographics — Age, Gender, Location, Subscription Status
 2. Purchase details — Item Purchased, Category, Purchase Amount, Season, Size, Color
 3. Shopping behavior — Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type, Payment Method
- **Missing Data:** 37 values in Review Rating (about 1% of rows)

3. Exploratory Data Analysis using Python

Data preparation and cleaning was done in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` and `.describe()` to check structure and summary statistics.
- **Missing Data Handling:** Filled missing Review Rating values using each product category's own median rating, preserving category-level differences.
- **Column Standardization:** Renamed columns to snake_case for consistency and easier use in SQL.
- **Feature Engineering:** Created `age_group` (quartile bins) and `purchase_frequency_days` (converted from text frequency labels).
- **Data Consistency Check:** Verified `discount_applied` and `promo_code_used` were identical in every row; dropped `promo_code_used`.
- **Database Integration:** Connected to MySQL via SQLAlchemy and loaded the cleaned dataset for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

Structured analysis was performed in MySQL to answer 14 business questions:

1. **Revenue by Gender** — Male customers generate \$157,890 in revenue vs. \$75,191 for female customers — more than double.
2. **High-Spending Discount Users** — 839 customers used a discount and still spent above the \$59.76 average purchase amount.
3. **Top 5 Products by Rating** — Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), and Skirt (3.78) — ratings cluster tightly, no standout favorite.
4. **Shipping Type Comparison** — Express shipping customers spend \$60.48 on average vs. \$58.46 for Standard — a modest, directionally sensible gap.
5. **Subscribers vs. Non-Subscribers** — Subscribers spend \$59.49 on average (1,053 customers, \$62,645 total) vs. \$59.87 for non-subscribers (2,847 customers, \$170,436 total) — subscribers do not spend more.
6. **Discount-Dependent Products** — Hat (50.0%), Sneakers (49.7%), Coat (49.1%), Sweater (48.2%), and Pants (47.4%) have the highest share of discounted purchases.
7. **Customer Segmentation** — Loyal: 3,116 customers (79.9%), Returning: 701 (18.0%), New: 83 (2.1%), based on previous purchase count.
8. **Top 3 Products per Category** — Leaders include Jewelry (Accessories), Blouse and Pants (Clothing, tied), Sandals (Footwear), and Jacket (Outerwear). Outerwear has only 2 distinct products in the dataset, so no 3rd-ranked item exists there.
9. **Repeat Buyers & Subscriptions** — Among customers with more than 5 previous purchases, only 27.6% subscribe (958 of 3,476) — essentially the same as the 27% subscription rate across all customers. Being a repeat buyer does not increase the likelihood of subscribing.
10. **Revenue by Age Group** — Young Adult (\$62,143) leads, followed by Middle-aged (\$59,197), Adult (\$55,978), and Senior (\$55,763) — an 11% spread from highest to lowest.
11. **Review Rating vs. Loyalty & Spend** — Essentially flat: customers rating below 3 spend \$58.51 on average vs. \$60.52 for those rating 4 and above — rating does not meaningfully predict spend or repeat purchases.
12. **Discount Usage vs. Loyalty & Spend** — Discount users spend slightly less (\$59.28) than non-users (\$60.13) on average — discounting shows no positive return on spend.
13. **Seasonal Pattern** — Fall leads with \$60,018 in revenue and \$61.56 average spend; Summer trails at \$55,777 and \$58.41 — roughly an 8% gap, the strongest signal among all factors tested.
14. **Payment Method vs. Loyalty & Spend** — Flat across all six methods, from \$58.95 (Venmo) to \$60.92 (Debit Card) — payment method has no meaningful effect on spend.

A cross-tab of discount usage by gender (built alongside this analysis) found that 100% of discounts were applied to male customers and 0% to female customers — the sharpest pattern in the dataset, and the basis for the first recommendation below.

5. Dashboard in Power BI

An interactive, one-page executive dashboard was built in Power BI to present these findings visually — five KPI cards, six charts covering category, season, age group, payment method, and subscription status, and a dedicated chart highlighting the discount-by-gender finding.



6. Business Recommendations

- **Audit the Discount Program** — discounts show no positive effect on spend or loyalty (finding 12) and are applied exclusively to male customers, who already generate over double the revenue of female customers (finding 1). Recommend testing discounts on female customers and reviewing spend on male-only discounting where it isn't earning a return.
- **Redesign Loyalty Around Actual Behavior** — subscription status doesn't correlate with higher spend (finding 5) or a higher chance of being a repeat buyer (finding 9), so the current subscription flag isn't functioning as a loyalty lever. Recommend tiered rewards based on real purchase history (the Loyal/Returning/New segments from finding 7) instead.
- **Lean Into Fall and Clothing** — Fall is the only season with a real revenue lift (finding 13), and Clothing is the leading category by a wide margin. Concentrate seasonal marketing, promotions, and inventory around this combination.
- **Targeted Marketing** — Young Adults are the top-spending age group (finding 10) and Express-shipping customers spend more on average (finding 4). Focus acquisition and retention campaigns on these two segments, where the data already shows above-average value.